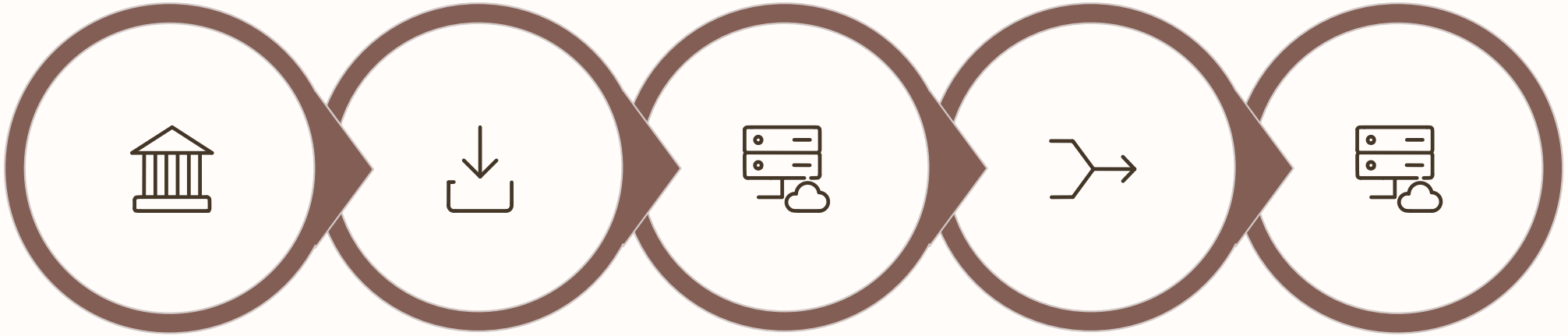
Olist Data Analysis Project

A comprehensive analysis of the Olist e-commerce platform using Python, covering 9 datasets: Customers, Products, Categories, Geolocation, Orders, Order Items, Sellers, Payments, and Reviews.

Made with GAMMA

Project Workflow



Import Libraries → **Load Data** → **Explore Data** → **Merge Tables** → **Clean & Analyze**

Data was merged in multiple stages — each followed by cleaning and EDA. A custom `Full_Data_Quality_Report` function checked missing values, data types, outliers, and duplicates.

Made with GAMMA

Revenue by Category

TOP PERFORMER

health_beauty leads with $1.44M

watches_gifts and bed_bath_table round out the top 3. Mid-range: sports_leisure and computers_accessories. Underperformers: garden_tools, auto, and housewares need stronger growth strategies.

Invest

Double down on health_beauty marketing

Diversify

Expand mid-range category offerings

Bundle

Cross-category promotions to lift underperformers

Made with GAMMA

Delivery Efficiency & Seasonal Revenue

Order Status Distribution

Over **110,000 orders** were successfully delivered. Cancellation and lost order rates are very low — confirming Olist's logistics system is robust and reliable.

Monthly Revenue Peaks

May peaks at ~$1.74M, while September drops sharply to ~$720K. Winter months are below peak but relatively stable. Boost spring campaigns and analyze fall declines to address seasonality.

Made with GAMMA

Product Weight & Freight Value

Positive Correlation

Heavier products incur higher freight costs, but at a **diminishing rate** — suggesting economies of scale in shipping.

Optimize Packaging

Reduce unnecessary weight without compromising safety. Negotiate freight contracts around weight thresholds for volume discounts.

Data-Driven Pricing

Use freight cost data to set accurate pricing. Monitor anomalies regularly to identify cost-saving opportunities.

Made with GAMMA

Customer Loyalty Depth

Sharp Drop-Off After 2nd Purchase

2,670 customers made 2 purchases; only **192** made 3. Super-loyal customers (9+ purchases) are rare exceptions.

→ **Tiered Loyalty Programs**

Progressive rewards to encourage continued engagement

→ **Personalized Marketing**

Target post-2nd-purchase customers with tailored offers

→ **VIP Treatment**

Nurture super-loyal customers to maximize lifetime value

Made with GAMMA

Geographic Revenue & AOV Analysis

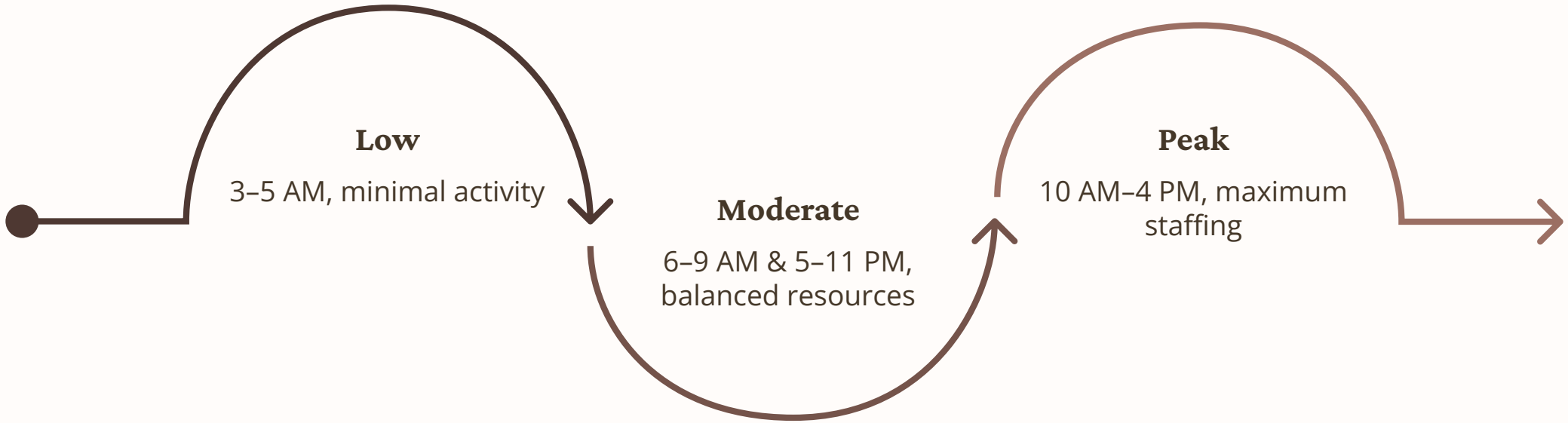
São Paulo dominates with $5.92M (~50% of top 10)

RJ and MG are key secondary markets. Notably, **Bahia has the highest AOV at $182** despite ranking 6th in revenue — distant states tend to bulk-buy due to higher shipping costs.

Strategy: SP → focus on retention & frequency. BA & GO → upsell premium bundles to high-AOV customers.

Made with GAMMA

Hourly Order Patterns



Peak hours (10 AM–4 PM) require maximum staffing and inventory readiness. Launch promotions during off-peak evening hours to stimulate demand and balance operational load.

Made with GAMMA

Payment Methods & Installment Behavior

Preferred Payment Methods

86.7K

Credit Card

Dominant method by far

22.9K

Boleto

Strong second preference

6.3K

Voucher

Limited usage

Installment Insights

58,599 orders paid in 1 installment (full payment preferred). A notable spike at **10 installments (6,842 orders)** — likely tied to a popular "Easy Pay" promotion. Payments beyond 10 are nearly non-existent.

Highlight top choices (1, 2, 3, 10) at checkout to streamline the user experience.

Made with GAMMA

Key Takeaways

Revenue Opportunities

health_beauty leads; seasonal peaks in May demand targeted campaigns

Logistics Strength

110K+ successful deliveries; optimize freight pricing by weight thresholds

Customer Retention

Loyalty depth is shallow — tiered programs and personalization are critical

Geographic Strategy

SP drives volume; BA & GO drive value — tailor strategies accordingly

Made with GAMMA